



LABORATORY MANUAL

**CE2006/CZ2006/CE207/CSC207: Software Engineering
SWLAB3 (Location: N4-B1C-14)**

**SCHOOL OF COMPUTER ENGINEERING
NANYANG TECHNOLOGICAL UNIVERSITY**

Overview

1. **INTRODUCTION**

The laboratory sessions form an integral part of the student's learning in this course.

Students will put into practice concepts and skills learnt in the lectures and tutorials, in the context of a real-world problem. They will solve the given problem with tools and practices commonly used in industry.

They will traverse the entire Software Development Life Cycle to deliver a medium-size project in a team. Students will contribute to different types of SDLC work products e.g. requirement, design, code, test cases.

The project is to be scoped such that it can be completed by a team of 4~6 students within 10 weeks.

2. **SOFTWARE TOOLS**

The following tools are available for use in the lab. Students must prepare their development environment accordingly. The lab technical executives are able to help with installation and configuration issues. Students may use other software tools that they deem relevant to their projects. However, there is no support for these tools.

<i>Name</i>	<i>Description</i>
GitHub	Source code management platform for developers https://github.com/
MySQL 5.5 (Community Edition)	Database Server
Java SE 1.7	JDK (Java Development Kit)
Netbeans 7.2	IDE (Integrated Development Environment)
Eclipse	IDE (Integrated Development Environment)
Visual Paradigm 10	UML Modelling Tool
Android SDK	adt-bundle-windowsx86_64-20140321
Scantist DevSecOps Platform	Development Security Operations emphasizes integrating security into every phase of the development lifecycle through collaboration, automation, and a culture of shared responsibility.

3. **ROADMAP**

The lab sessions will guide students in building the system step-by-step, culminating in a demo where students will showcase their work.

	<i>Lab Focus</i>	<i>Deliverables</i>
Lab #1	Requirements Elicitation (2 Weeks)	• Functional Requirements • Non-Functional Requirements • Data Dictionary • UI Mockups • Initial Use Case Model
Lab #2	Requirements Analysis (2 Weeks)	• Complete Use Case Diagram • User Case Descriptions • Class Diagram of Entity Classes • Identification of Major Boundary Classes and Control Classes • Sequence Diagrams of some use cases • Initial Dialog Map (State Machine of System UI)
Lab #3	Design and Implementation (5 Weeks)	• Complete User Case Descriptions • Complete Class Diagrams • Complete Sequence Diagrams • Complete Dialog Map • System Architecture • Application Skeleton
Lab #4	Implementation, Testing, and Preparing for Demo (2 Weeks)	• Test Cases and Testing Results • Working Application • Demo script
Lab #5	Project Demo and Submission (Week12 or Week13)	• Live demo • Final documentations • Peer reviews

In Lab#1, your Supervisor will assign you into teams of 4-6 people, including a team leader. The teams, once formed, shall not be changed unless there is a need due to changes during the add/drop period or other considerations agreed by the Lab Supervisor.

Lab activities will mainly happen in between lab sessions. Your team should work together to produce the required deliverables.

During the Lab#2 (Lab#3 or Lab#4), your lab supervisor will evaluate your project progress against the required deliverables of the Lab#1 (Lab#2 or Lab#3).

Final documentations and peer reviews are due on 23:59:59, ending Sunday of Week 12 for the teams who demo in Week12, or 23:59:59, ending Sunday of Week 13 for the teams who demo in Week13.

4. **ASSESSMENT**

Team mark

- Continuous assessment: 15% (5% each for Lab#1/Lab#2/Lab#3 deliverables)
- Live demo: 15% (must be done in the Lab#5)
- Final documentations: 15% (due on Sunday midnight **after** your demo)

Individual mark

- $\text{team_mark} \times \text{individual_contribution_weight}$ determined by Peer and Supervisor assessment

Continuous assessment

- Ensure that the teams follow the SDLC steps properly and make good progress towards the final demo
- Correctness of the deliverables is **NOT** the focus of the continuous assessment. Deliverables can still be revised in the subsequent lab sessions.
- Use three levels of criteria
 - Perform the required SDLC activities and produce a clear collection of all the required deliverables
 - Start the required SDLC activities and produce some required deliverables
- Please submit the required lab deliverables to your dropbox or GoogleDrive repository before the lab starts.

Live demo

- Demonstrated understanding of the SDLC activities throughout the project
- Demonstrated system functionalities against requirements specification

Final documentations

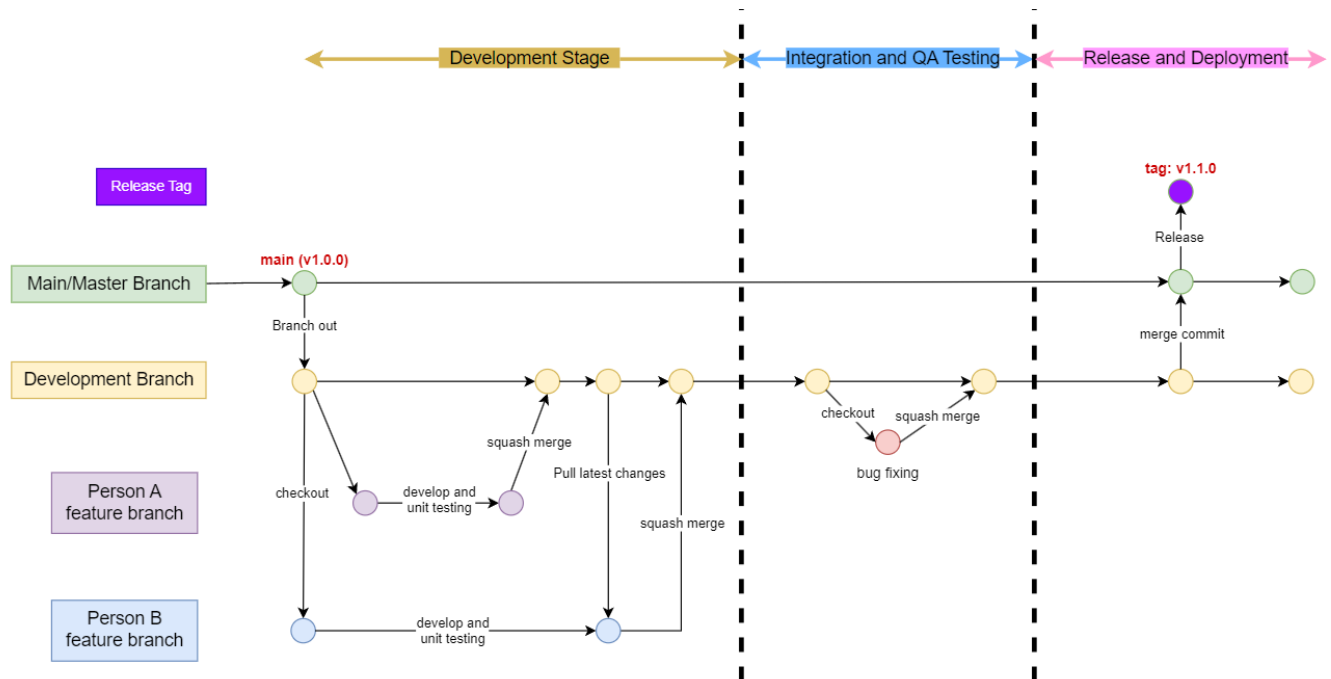
- Completeness and correctness of the final documentations
- Effective use of good software engineering practices, architecture, design

Individual mark

- $\text{individual_contribution_weight}$ will be calculated based on peer reviews

APPENDIX – Using Github as one Team

For any repository, if there are multiple developers contribute and maintain the changes as one team, the following best practices shall be applied for the common SDLC workflow.



- Every repository by default shall have at least two branches:
 - main/master: considered the stable branch of a repository which represents the latest stable release or production-ready code.
 - dev: served as an integration branch for on-going development works including various features, bug fixes, and other changes.
- Every developer shall create (from the dev base branch) and work on their individual feature branches for any changes they are going to make. Similar to bug fixing branch as well.
- It's important to pull latest code from the base again before merging any changes back to base development to avoid any changes happened and commit by other contributors.
- After all development and testing activities, the final version of development branch need to merge back to main/master branch and also it's important to tag your latest version.

Description	Git commands
Switch to main development branch	git checkout dev
Create feature branch from the base development branch	git checkout -b featureABC
Raise commit to local repository	git commit -m "commit message"
Push commit to remote repository	git push
Check local commit history	git log
Pull latest changes from remote base development branch	git pull origin featureABC
Combine multiple commits into single one and merge into target branch	git merge --squash dev

APPENDIX – Using DevSecOps Platform

1. Sing up

To start using DevSecOps platform, sign up for an account by Linking your app account with your GitHub accounts directly.

- a. Go to platform login page: <https://app.scantist.io/>
- b. Click on Github icon
- c. Enter your Github credentials to authorize Scantist
- d. You will be redirected to platform and logged in automatically

Once you have linked your GitHub account with Scantist, you will be able to scan your open source projects directly from your GitHub repositories and pipelines. This provides a more streamlined process for managing and securing your open source software or applications.

2. Create a project

To begin scanning your project for vulnerabilities, you'll need to create a project:

- a. Go to “Projects” page from menu bar on the left
- b. Choose “SCM Platforms” under “Create Project” button
- c. The prompt page will link to your GitHub and list down all authorized repositories.
- d. By clicking Add button, the repository contents will be imported into platform and create project accordingly.
- e. During project creation, you could also indicate whether you would like to trigger new scan immediately after project creation.

3. Initiate a scan on your project

You can trigger a new scan for your project and select two available engines:

- Software Composition Analysis (SCA)

Identify third-party components and their associated vulnerabilities or license risks.

- Static Application Security Testing (SAST)

Provide static analysis of application source code for security vulnerabilities and issues.

4. Access your scan results

After scanning your projects, you can view scan results inside project detail page.

You could also download various format of reports in the page too.